EUROSAT MODEL ADAPTATION STUDY

Benchmark, fine-tune, and export lightweight vision models for aerial imagery.

This project benchmarks modern efficient backbones on EuroSAT, adapts the strongest models with supervised transfer learning, and packages the best practical candidate as an ONNX deployment artifact.

Head-reset transfer baseline

Recorded on NVIDIA GeForce RTX 3060 Laptop GPU

GitHub Repository [https://github.com/KenanTrivedi/efficient-vision-benchmark]

Project PDF [assets/EuroSAT_Edge_Vision_Benchmark_Portfolio.pdf]

README [https://github.com/KenanTrivedi/efficient-vision-benchmark/blob/main/README.md]

PROJECT FLOW

1. Validate the local EuroSAT cache and benchmark the corrected 10-class split.
2. Fine-tune the strongest candidate models with a staged transfer recipe.
3. Export the best practical model as FP32 and INT8 ONNX artifacts.

MODELS BENCHMARKED

9

Architectures in the baseline sweep

BEST BASELINE ACCURACY

FastViT-MCI4

13.41 % on the held-out test split

FASTEST CPU MODEL

MobileNetV4-Small

15.62 ms median CPU latency

TOP FINE-TUNED MODEL

ConvNeXt-Tiny

99.14% top-1 after supervised adaptation

METHODOLOGY

Interpret the baseline correctly

Each pretrained backbone is adapted to EuroSAT by replacing the final classifier with a fresh 10-class layer and evaluating before training.

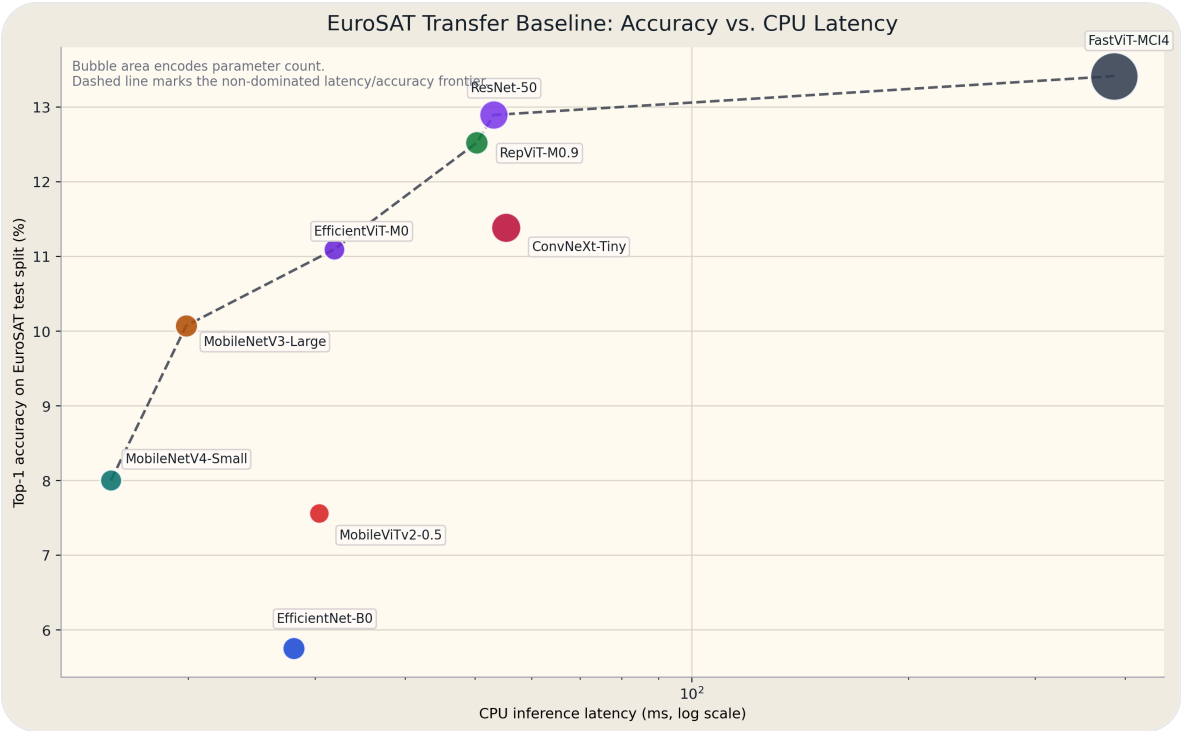
WHAT CHANGED IN THIS PROJECT

From screening run to publishable pipeline

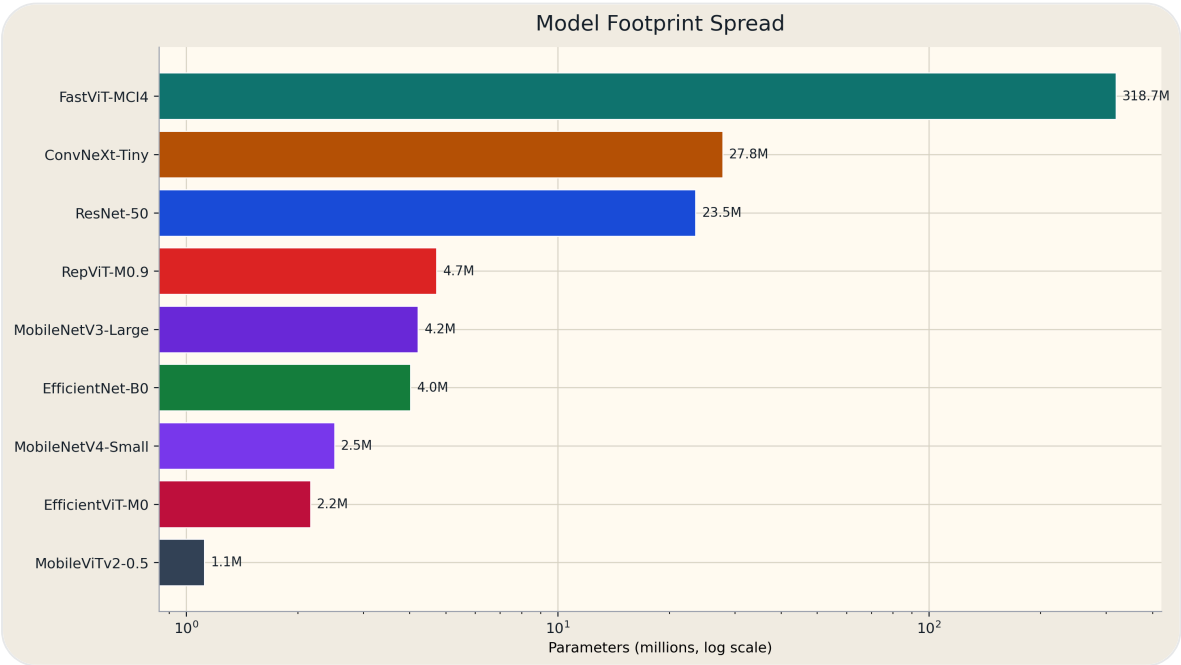
- The dataset cache is checked before the run so incomplete EuroSAT copies are rejected early.
- The fine-tuning stage uses a staged recipe instead of a single fixed loop.
- The deployment stage shows only the artifacts that were actually generated and validated.

BENCHMARK FIGURES

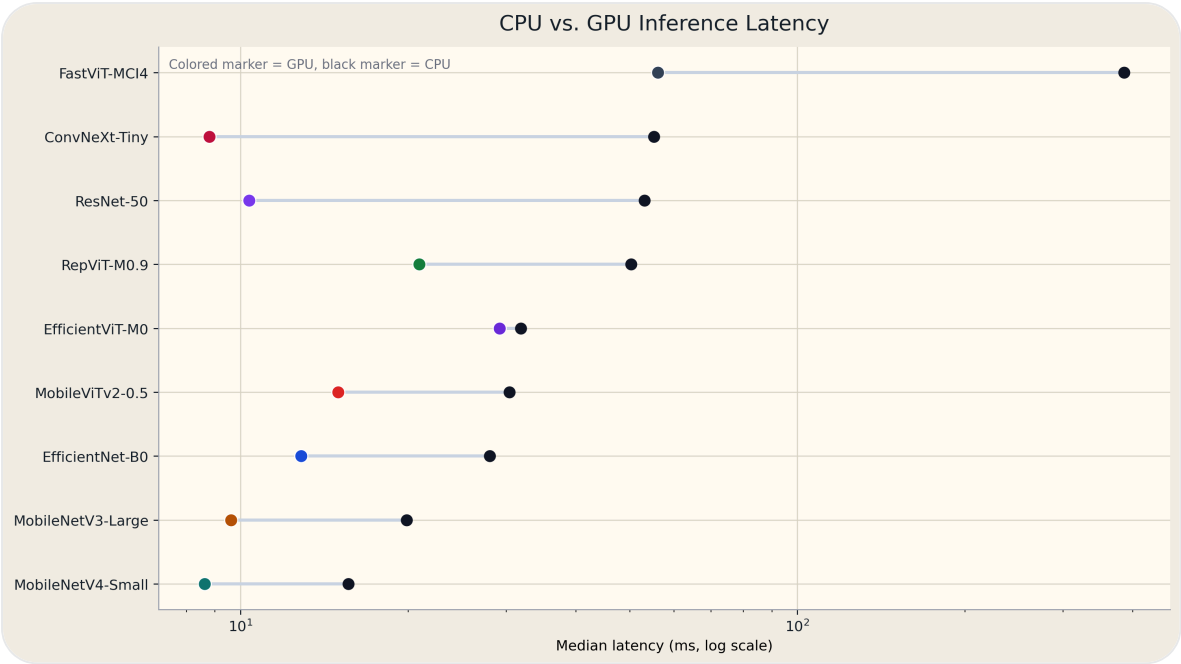
Latency, footprint, and adaptation behavior



Initial screening step: accuracy versus CPU latency with a Pareto frontier and bubble sizes scaled by parameter count.



Log-scale footprint plot to make size differences legible across small and very large backbones.



CPU versus GPU latency breakdown for the same baseline models.

BENCHMARK LEDGER

Normalized baseline view

SORTED BY CPU LATENCY

Model	Top-1	CPU	GPU	Params	MACs	Batch-1
MobileNetV4-Small	8.00%	15.62 ms	8.62 ms	2.5M	188M	138.7 img/s
MobileNetV3-Large	10.07%	19.87 ms	9.62 ms	4.2M	230M	97.7 img/s
EfficientNet-B0	5.75%	28.02 ms	12.84 ms	4.0M	409M	75.8 img/s
MobileViTv2-0.5	7.56%	30.39 ms	14.98 ms	1.1M	472M	81.7 img/s

Model	Top-1	CPU	GPU	Params	MACs	Batch-1
EfficientViT-M0	11.09%	31.88 ms	29.19 ms	2.2M	79M	41.2 img/s
RepViT-M0.9	12.52%	50.31 ms	20.95 ms	4.7M	839M	57.2 img/s
ResNet-50	12.89%	53.10 ms	10.36 ms	23.5M	4130M	114.6 img/s
ConvNeXt-Tiny	11.38%	55.22 ms	8.80 ms	27.8M	4488M	101.7 img/s
FastViT-MCI4	13.41%	385.84 ms	56.13 ms	318.7M	22112M	16.5 img/s

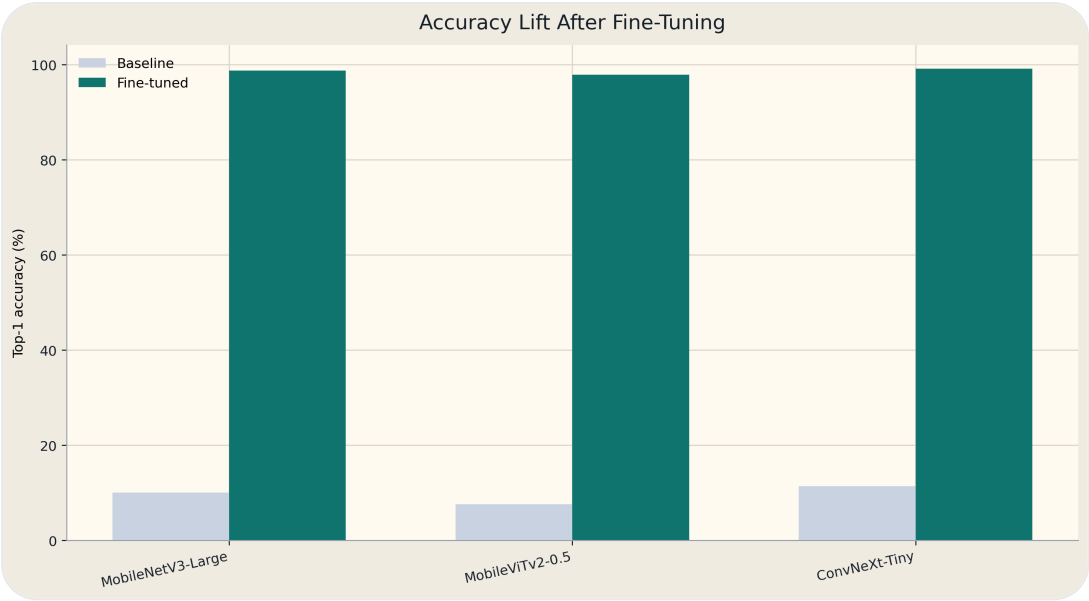
FINE-TUNING OUTCOME

Supervised adaptation on EuroSAT

DEPLOYMENT TARGET: MOBILENETV3-LARGE — FASTEST MODEL WITHIN 1 PERCENTAGE POINT OF THE TOP FINE-TUNED ACCURACY.

3 fine-tuned models completed. Supervised adaptation closes the transfer gap and produces a practical export candidate.

Model	Top-1	Gain
ConvNeXt-Tiny	99.14%	+87.76 pts
MobileNetV3-Large	98.74%	+88.67 pts
MobileViTv2-0.5	97.90%	+90.34 pts

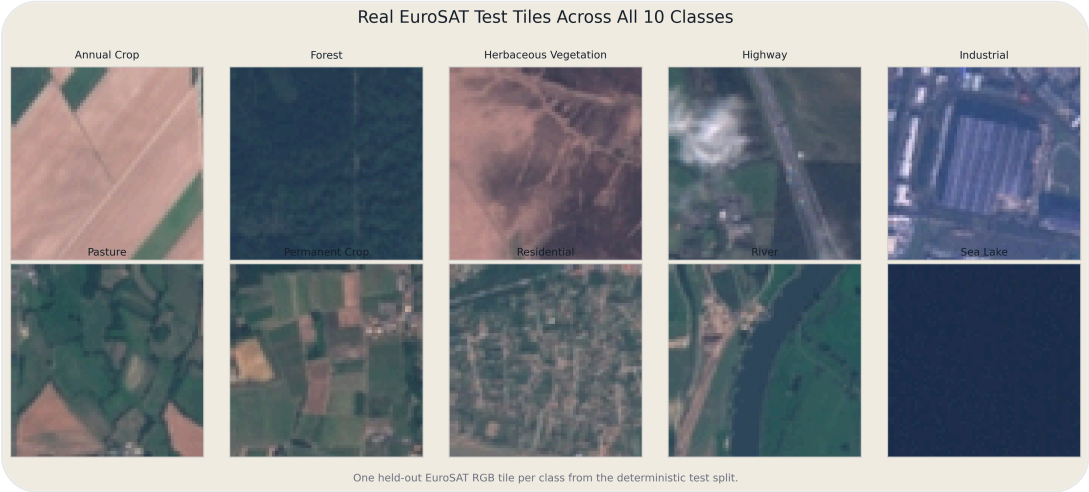


Baseline versus fine-tuned accuracy on the held-out EuroSAT test split.

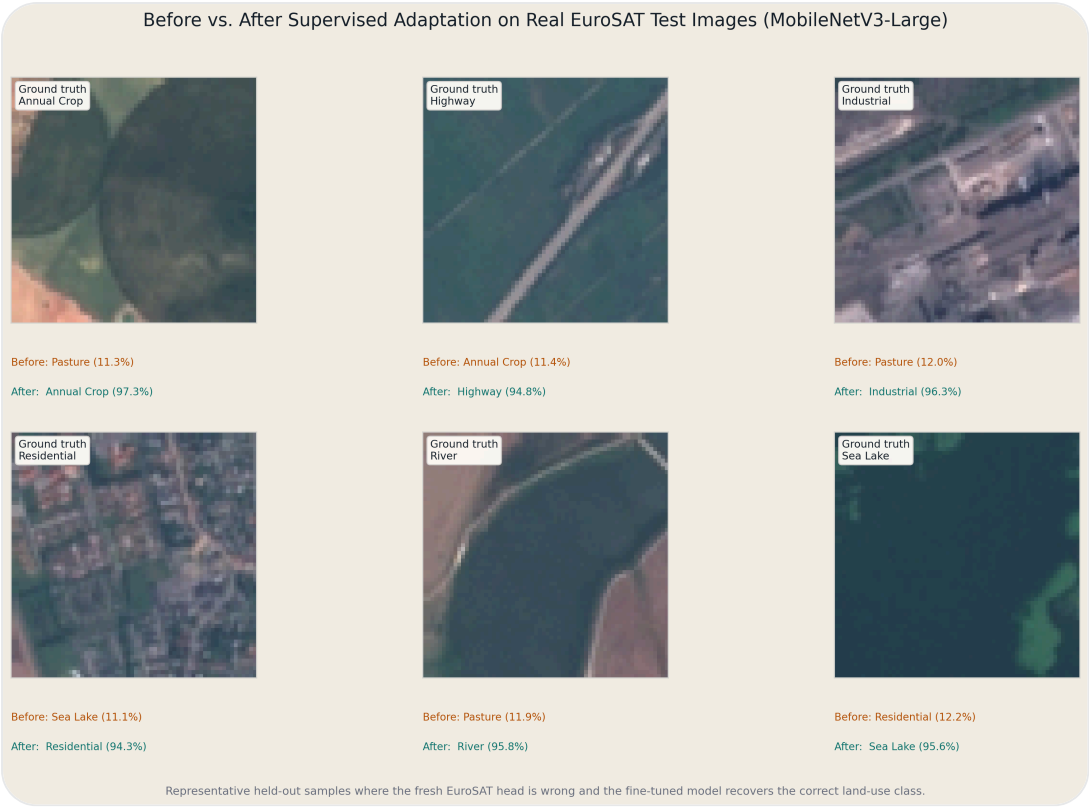
QUALITATIVE CHECK

Real EuroSAT imagery before and after adaptation

Real held-out EuroSAT test imagery. The dataset panel shows one test tile per class, and the before/after panel uses MobileNetV3-Large to show how supervised adaptation corrects the fresh 10-class head on representative aerial samples.



One held-out EuroSAT test tile per class from the corrected 10-class split.



Representative real test tiles with predictions before and after supervised adaptation.

SELECTED MODELS

Fine-tune seeds from the baseline sweep

1. MobileViTv2-0.5 - selected as a supervised adaptation candidate.
2. MobileNetV3-Large - selected as a supervised adaptation candidate.
3. ConvNeXt-Tiny - selected as a supervised adaptation candidate.

EXPORT DECISION

How the deployment candidate was chosen

The export pipeline packages MobileNetV3-Large into validated FP32 and INT8 ONNX artifacts, together with reproducible calibration data for the quantization step.

COMPLETED EXPORTS

Artifacts generated from the selected model

MOBILENETV3-LARGE • 4 GENERATED DEPLOYMENT ARTIFACTS

AVAILABLE

FP32 ONNX

results/deployment/mobilenet_v3_large/mobilenet_v3_large_fp32.onnx • 16.07 MB

VALIDATED

ONNX validation

max abs diff 0.003182 • mean abs diff 0.001391

AVAILABLE

Calibration data

results/deployment/mobilenet_v3_large/calibration_data.npz • 256 samples • 147.00 MB

AVAILABLE

INT8 ONNX

results/deployment/mobilenet_v3_large/mobilenet_v3_large_int8_qdq.onnx • 4.25 MB • onnxruntime-static-qdq

PROJECT LINKS

Source, page, and PDF

CODEBASE

GitHub repository

Python scripts, site files, and generated summaries.

[https://github.com/KenanTrivedi/efficient-vision-benchmark]

PROJECT PAGE

GitHub Pages site

The published version of this page with the current figures.

[https://kenantrivedi.github.io/efficient-vision-benchmark/]

PDF

Printable summary

A PDF export of the project page with visible hyperlinks.

[assets/EuroSAT_Edge_Vision_Benchmark_Portfolio.pdf]

EuroSAT • train 18900, val 4050, test 4050 • generated 14/03/2026, 23:04:21

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